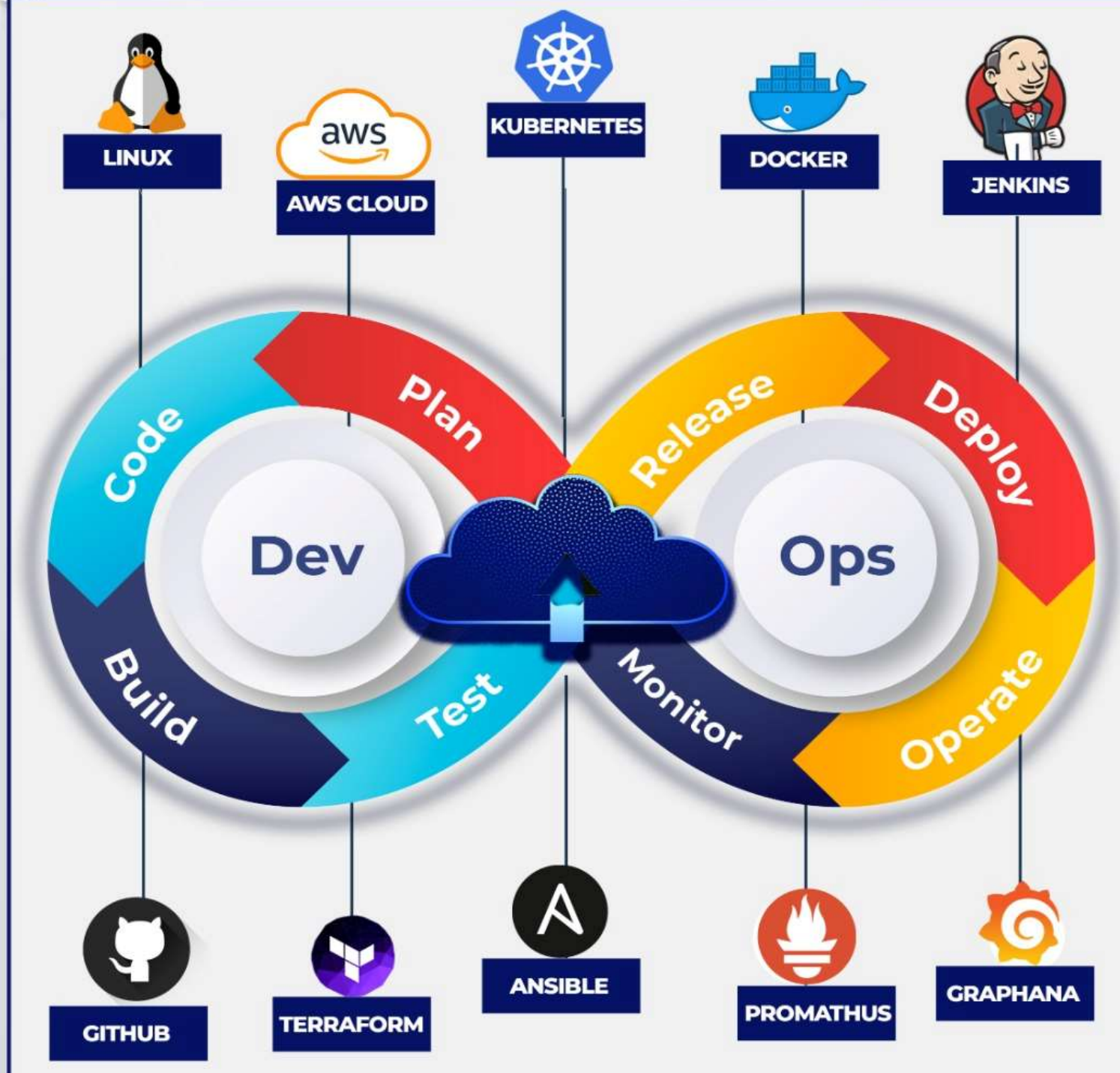
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DevOps & Cloud Computing



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ABOUT US

UpFlairs is an innovative educational technology company with a clear mission to elevate the skills and employability of students throughout India.

Our dedicated team is committed to fostering the next generation of tech talent, equipping them with cutting-edge skills in emerging technologies and has educated more than 47K students all over the globe including IITs, NITS, Deemed Universities and other colleges.

We offer the courses that are most trending technologies of the recent era in the fields like AI/ML, Data Science, Cloud Computing, DevOps, Full Stack Web Development, Embedded Systems, IoT, and Robotics. These courses are meticulously designed to provide students with the practical skills and knowledge required to excel in tech-driven careers, making them not just job-ready but industry leaders of tomorrow.

We are not only limited to training the youth of the country but also provide Lab setups to various Colleges and Universities and Project solutions to other companies for the domains like AI-ML, IoT, ROBOTICS AND CLOUD.

EDUCATION



PROJECT SOLUTION



LAB SETUP



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CAREER OPPORTUNITIES IN DEVOPS & CLOUD COMPUTING

Key Job Roles in DevOps & Cloud Computing

- DevOps Engineer
- Cloud Architect
- Site Reliability Engineer (SRE)
- Cloud Engineer
- DevOps Consultant
- Cloud Networking Engineer

Industries Hiring DevOps & Cloud Computing

- IT & Software Development
- Finance & Banking
- Healthcare
- E-commerce
- Telecommunications
- Media & Entertainment

Benefits of Career in DevOps & Cloud Computing

- High Demand & Job Security
- Attractive Salary Packages
- Fast Career Growth
- Versatile Skill Set
- Collaboration Opportunities
- Remote Work Opportunities

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MODULE HIGHLIGHT



COMPREHENSIVE CURRICULUM

Learn DevOps and Cloud Computing with in-depth modules on technologies like Linux, AWS Cloud, Kubernetes, Docker, Jenkins, and more.

INDUSTRY PROJECTS

Gain practical experience by working on real-world projects that mirror industry scenarios.





CAREER COACHING

Access resume-building sessions, interview preparation, and personalized career guidance.

MOCK INTERVIEWS & TEST SERIES

Prepare with regular mock interviews & technical test series to enhance your job readiness.





DAILY ASSIGNMENTS & MENTORSHIP

Strengthen your learning with daily assignments and 1:1 mentorship from industry experts.

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DEVOPS & CLOUD COMUTING

Duration – 6 Months (240 Hours)

Module 1: Linux Fundamentals (20 Hours)

INTRODUCTION (2hr)

- What is linux?
- Why linux?

INSTALLATION (2 hr)

- Installing Centos / Redhat / Ubuntu
- Virtualization
- Virtual box

BASIC COMMANDS (4hr)

- What is commands
- Linux system commands
- System Utility Commands in Linux (date,uptime, hostname, uname, which, cal, bc)

TEXT FILE EDITORS (2hr)

- A Gentle Introduction to vi , vim ,nano

INPUTOUTPUT REDIRECTION (2hr)

- Standard input , standard output and standard error
- Redirecting Output to a file or Program
- Constructing pipelines

VARIABLES (3hr)

- The Environment
- Shell variable
- Special variable ($0 ,$$,$?)
- Customizing the Prompt

ARCHIVING AND BACKUP (1hr)

- Managing Compressed Tar Archives
- Extracting files of an archive
- Create a compressed file

USERS AND GROUPS(2hr)

- User and Group Concepts
- Gaining superuser access
- Manage local users
- Manage local Groups
- Manage Users Password

PERMISSION (2hr)

- Basic permission
- Read
- Write
- Excute

BASIC NETWORKING (2hr)

- Networking concept
- DHCP & DNS
- Webserver apache2

Module 2: Linux Administration (18 Hours)

ADVANCED COMMANDS (2hr)

- Systemctl command
- ps command
- top command
- kill command
- Process Management (bg, fg, nice)

SPECIAL PERMISSION (2hr)

- Octal of basic permission
- Access control list
- SUID
- GID
- sticky bit

TASK SCHEDULING AND CRON JOBS (1hr)

- Introduction to Task Scheduling
- Understanding Cron Jobs
- Creating and Managing Cron Jobs
- Practical Examples of Cron Jobs
- Automating Repetitive Tasks with Cron

SHELL SCRIPTING (3hr)

- Writing Your First Script
- Variables in Shell Scripting
- Loops
- If-Else Statements
- Operators

MONITORING AND MANAGING LINUX PROCESS (2hr)

- Processes
- Killing process
- Monitoring process Activity

STORAGE (3hr)

- Storage and partition using GPT and MBR
- LVM and thin provision
- Swap memory

ANALYZING AND STORING LOGS (2hr)

- Describing System Log Architecture
- ReviewingSyslog Files

ADVACNEC NETWORKING (3hr)

- Network profiling and nmcli
- Firewalld and iptables
- NFS
- SSH & RDP

Mini Project 1: (3 hrs)

- ***Creating a command-line-based chatting platform using Netcat***
- ***Building a Basic Linux Monitoring Script Using Shell***

Module 3: Cloud Platform with AWS (15 Hours)

INTRODUCTION (2hr)

- What is cloud computing
- Types Of Cloud Computing
- Cloud Computing Issues
- Virtualization
- Cloud Delivery Models
- Features of Cloud Computing
- Amazon Web Services (AWS)
- Introduction To AWS
- Introduction To The AWS Products, Regions and Availability Zones
- Signing Up For AWS
- Introduction AWS Management Console

ELASTIC COMPUTE CLOUD (EC2) (5hr)

- Introduction to EC2 and Storage
- Amazon Machine Images (AMI)
- Instance Family General Purpose, Compute, Memory, Storage
- Placement Group
- AWS Pricing Model

ELASTIC LOAD BALANCING AND AUTO SCALING (2hr)

- Introduction to Elastic Load Balancing
- Types of Load Balancers
- Application load balancer
- Network load balancer
- Auto Scaling

AWS IAM (2hr)

- Introduction to AWS IAM
- IAM Users and Groups
- IAM Roles
- IAM Policies
- IAM MFA (Multi-Factor Authentication)

DATABASES (4hr)

- AWS Databases - Introduction, Benefits, and Types
- Relational Database Service (RDS) - Features and Benefits
- Mysql Serverless Relational Database

Mysql - Primary Key, Partitions, and Features

Module 4: AWS Essential Services (20 Hours)

SIMPLE STORAGE SERVICE(S3) (4hr)

- Storage services – EBS , S3 & EFS
- S3 Introduction - Durability, Consistency, Usage, Object Ownership
- S3 Storage Classes
- S3 Versioning

MONITORING RESOURCES IN CLOUD (3hr)

- Introduction to Monitoring Services
- Monitoring with cloud watch
- CloudWatch, Metrics and Alarms
- CloudWatch Logs - Consolidate and Monitor Logs
- Lab - customising web server log on cloudwatch

VIRTUAL PRIVATE CLOUD (VPC) (3hr)

- VPC deep dive
- Your VPCs
- New
- Subnets
- Route tables
- Customer gateways
- Virtual private gateways
- Site-to-Site VPN connections

AWS ELASTIC BEANSTALK (3hr)

- Introduction to Elastic Beanstalk
- Overview of Elastic Beanstalk
- Benefits and Use Cases
- Deployment Models
- Supported Platforms (e.g., Node.js, Python, Java, .NET, etc.)
- Deployment Options (Single Instance, Load Balanced, Auto Scaling)

AWS LAMBDA (3hr)

- Introduction to AWS Lambda
- Understanding the Lambda Execution Model
- Creating Your First Lambda Function

AWS CONTAINER SERVICES (2hr)

- Aws elastic container services
- Benefits and features
- Pricing

AWS KUBERNETES SERVICES (2hr)

- Introduction of kubernetes

Mini Project 2: (5 hrs)

- ***"Deploying two tier application using amazon ELB"***
- ***"Deploy project on elastic beanstalk"***

Module 5 : Foundations of Version Control (6 Hours)

- Subversion Controls/Git
- Working with local repositories
- Working with remote repositories
- Branching

Module 6: Expert GIT and Subversion (6 Hours)

- Merging
- Cloning
- Fetch/pull
- Rebase and tag
- Collaboration
- Multi project

Module 7: Containerization Platforms (30 hours)

INTRODUCTION (2 hr)

- ContainerHistory
- Container runtime engines
- Docker architecture understanding

INSTALLATION OF DOCKER ON VARIOUS PLATFORMS (3hr)

- Installation Methods for Docker
- Installation in linux
- Installation docker-desktop

DOCKER OPERATIONS (4hr)

- Basic command of docker
- Docker Image vs Docker Containers
- Container Identification
- Port Binding
- Attached and Detached Modes
- RemovingDocker Containers
- docker container exec
- Importance of IT flags

IMAGE CREATION, MANAGEMENT, AND REGISTRY (3hr)

- Docker Commit
- Working with Docker Images
- Overview of Dockerfile

- COPY vs ADD Instructions
- Dockerfile
- Dockerfile - ENTRYPOINT Instruction
- Dockerfile - WORKDIR Instruction
- Dockerfile - ENV Instruction
- Document - ENV Instruction Code
- Tagging Docker Images
- Python and java based containerization
- Layers of Docker Image
- Managing Images with CLI
- Inspecting Docker Images
- PruningDocker Images
- Overview of Docker Registries
- Pushing image to dockerhub
- Pushing image to ECR and ACR

NETWORKING (4hr)

- Overview of Docker Networking
- Understanding Bridge Networks
- Implementing User-Defined Bridge Networks
- Understanding Host Network
- Implementing None Network

STORAGE AND VOLUMES (4hr)

- ChangingStorage Drivers
- Overview of Docker Volumes
- Bind Mounts
- DOCKER WEBUI (2hr)
- Introduction of portainer
- Demo using portainer

DOCKER COMPOSE (4hr)

- Introduction
- Demo
- Compose multi stage example

ORCHESTRATION (4hr)

- Overview of Container Orchestration
- Overview of Docker Swarm & Building Labs

- Document - Docker Script
- Initializing Docker Swarm
- Services, Tasks and Containers

Mini Project 3: (3 hrs)

- ***"A multi-tier microservices testing using docker compose".***

Module 8: Kubernetes (42 hours)

INTRODUCTION (3hr)

- Why kubernetes
- History of kubernetes
- Kubernetes components discussion
- SETUPK8S CLUSTER (5hr)
- Single node using minikube
- Custom k8s
- Cloud k8s -EKS AKS GKE

COMMON KUBECTL COMMANDS (4hr)

- Creating,
- Updating, and
- Destroying Kubernetes Objects
- POD (4hr)
- Pod concept
- Pod details
- Multi-container pod understanding
- Access k8s APIs using curl

KUBERNETES NETWORKING AND KUBERNETES SERVICE (4hr)

- Introduction of CNM and CNI
- K8s service
- Nodeport service demo

HTTP LOAD BALANCING WITH INGRESS (4hr)

- Introduction of Ingress
- Configuring local DNS

DEMO CONTROLLER (4hr)

- Intro to replication controller
- Kubernetes workload according to apps
- Configmap demo

DEPLOYMENT (5hr)

- Intro to deployment
- Javabase app deployment

KUBERNETES SECRETS (3hr)

- Introduction of k8s secretes
- Listing ,creating, updating

KUBERNETES RBAC (3hr)

- Understanding Roles and Role Bindings
- Kubernetes RBAC demos

KUBERNETES HELM DEMOS (3hr)

- Introduction

Mini Project 4: (3 hrs)

- ***"Deploying a Database replica using stateful-sets"***

Module 9: Terraform (15 Hours)

INTRODUCTION(1 hr)

- History of Terrafor
- Installation and Setup

TERRAFORM CONFIGURATION AND HCL (4 hr)

- Setting up AWS Account
- Deploying Infrastructure with Terraform
- Terraform Code - First EC2 Instance
- Resource and Providers
- Terraform Destroy
- Understanding Terraform State File
- Understanding Desired & CurrentStates
- AWS Provider - Authentication Configuration

TERRAFORM VARIABLES AND OUTPUTS (3 hr)

- Read, Generate, Modify Configurations
- Output Values
- Terraform Variables
- Data Types for Variables
- Fetching Data from Maps and Lists in Variable
- Terraform Functions
- Terraform Output

TERRAFORM PROVISIONERS (2 hr)

- Understanding Provisioners in Terraform
- Types of Provisioners
- Implementing remote-exec Provisioners

- Implementing local-exec Provisioners

TERRAFORM MODULES AND WORKSPACE (2 hr)

- UnderstandingDRY Principle
- Implementing EC2 Module with Terraform
- Variables and Terraform Modules
- Using Locals with Modules
- Referencing Module Outputs

TERRAFORM REGISTRY AND AUTOMATION (1 hr)

- Terraform Registry
- Automation and Provisioning of AWS
- Advanced Terraform

Mini Project 5 : (2 hrs)

- ***"Launch a Webserver in EC2 instance with help of terraform"***

Module 10: Ansible (12 hours)

INTRODUCTION (1 hr)

- Why Ansible?
- Lab - Setup VMs and Install Ansible

ANSIBLE INVENTORY (2 hr)

- Introduction to Ansible Inventory
- Demo - Ansible Inventory

Introduction to YAML (2 hr)

ANSIBLE PLAYBOOKS (3 hr)

ANSIBLE VARIABLES (1 hr)

CONDITIONALS AND LOOPS (2 hr)

ANSIBLE ROLES (1hr)

Mini Project 6 : (2 hrs)

- ***"Real World Implementation"***

Module 11 : CICD- Jenkins (20 hours)

- What is ci &cd
- Need of the industry
- Installing Jenkins
- Understanding Jenkins
- Creating jobs and connecting to technology
- Docker connecting
- Aws connection

- Design entire pipeline.

Module 12 : Monitoring with PROMETHUS (7 hours)

PROMETHEUS (7 hr)

- What is Prometheus?
- Installing prometheus on ubuntu(Linux)
- Overview of Scraping and Push Gateway
- Data model of Prometheus
- Data Types in Prometheus

Module 13 : Monitoring with GRAFANA (7 Hours)

GRAFANA (7 hr)

- What is Grafana?
- The Shoe Hub Company
- Introduction to Grafan
- Installing Grafana on Ubuntu (Linux)
- Configuring Grafana
- Connect Grafana to Prometheus
- Creating Your First Grafana Dashboard
- The Time Series Panel
- Annotations in Grafana
- Alerts in Grafana
- User management in Grafana

Capstone Project 1 :

- ***"Book what you want"***
- ***"Design a microservice oriented container networking solution where there will be 3 tier models like fronted , backend and databases. Try to keep them in different network bridges."***
- ***"Deploy statefulset for creating multiple mysql pods by keeping this in mind that your storage provisioner must be running within a cluster and keep database architecture in mind."***
- ***"Using ansible create a working 3 node k8s cluster , use kubeadm inside ansible "***
- ***"Deploy a sample web application using SSL in k8s under nginx ingress controller."***
- ***"Design a terraform script which can provision and deploy highly available architecture"***

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TRAINING COMPLETION CERTIFICATE

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CERTIFICATE
OF PARTICIPATION

Proudly Presented to

Student Name

X Sem |Department of

from X College has successfully completed 6 Months Internship & Training Program in **Full Stack Web Development** from D /M / Y to D /M / Y at **Upflairs Pvt. Ltd.**

Siddharth Singh
Founder & Director
Upflairs Pvt. Ltd.

MSME
MICRO, SMALL & MEDIUM ENTERPRISES
OUR STRENGTH
Ministry of MSME, Govt. of India

#startupindia

Enrollment No. : UF/XXX/XXXX

Trainer Name
Designation
Upflairs Pvt. Ltd.

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OUR STUDENTS FEEDBACK

I'm very happiest to share my experience about my summer internship at Upflair was an incredible learning experience in the data science domain. The hands-on projects, supportive mentors, and real-world applications made it the perfect opportunity for college students to develop their skills. The team's dedication to helping us grow both technically and professionally truly stood out, making it a highly rewarding experience and it is really good to learn something valuable.

Ritu Saini
PIET, Jaipur

I had a good experience at learning in upflairs. We had regular classes and practical implementation of machine learning and data science projects....
Faculty is best as educator and also teaches everything in beginner friendly manner.
We did a many projects and practically learnt the deployment.

Riya Sharma
Arka Jain University

I recently completed my Python internship at Upflairs. The program was well-structured, offering a solid blend of theoretical knowledge and hands-on projects. The mentors were incredibly supportive, providing guidance whenever needed and encouraging creativity and problem-solving.

Sanjay kumar
Arya College of Engineering

It was a great and amazing experience in upflairs pvt ltd, In my 45 days intership of data science and machine learning & AI I have learnt a lot and it helped in making a strong base for data science and ML & AI .

Manvik
PIET, Jaipur

I Anshika Mundra from Poornima Institute of Engineering and Technology have completed my summer training cum internship of 15 days in Frontend Web Development Course. My experience here was fabulous .The knowledgeable and understanding instructors helped me gain various technical and practical skills. I worked on various projects, applying my knowledge to various tasks.

Anshika Mundra
PIET, Jaipur

Great learning experience at Upflairs . Gained valuable skills, worked on meaningful projects, and enjoyed a supportive team environment. Highly recommend for hands-on experience and professional growth. Minor challenges, but overall positive.

Hotri Shrimali
Amity University, Rajasthan

Hello everyone, My name is Antish kumar from career point university. I had enrolled in 2 months Data sciences internship + traning program. The staff is so friendly, helpful to students and resolve all doubts. Our sir Ranjit singh assist me or whole class very well.

Antish Kumar
Career Point University, Kota

It was a really great experience in learning about domain and all the concept of Full stack web development. Environment and level of teaching skills were really up to the mark. Overall, it was a really good experience.

Ajit Prajapat
PIET, Jaipur

I completed my 45-days internship with Er. Ranjit Singh in Upflairs in ye domain of data science with AI&Ml... Here I got a Vry gud xperience with full practical implementation in AI&ML .... The faculty taught all the concepts very well...including the coding part ..... I did multiple projects under his guidance and also developed my final project in the same domain .

Akash Mishra
Arka Jain University

GLANCE

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